What a radical-pair quantum reservoir would require: readable capacity, no quantum advantage, and a nuclear-register criterion

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(Dated: August 3, 2026)

The radical-pair mechanism, by which a field as weak as the Earth's alters a chemical reaction, underlies the leading model of avian magnetoreception. Whether the electron–nuclear spin coherence behind this sensing could also support biological *computation*—a "quantum brain"—has been conjectured for decades but never made quantitative. We treat a cryptochrome-type radical pair as a reservoir computer, solving the radical-pair master equation with electronic decoherence and driving it as a real turnover does—by the birth of a spin-correlated pair whose singlet character encodes the input. Three results follow, none of them the expected one. First, the reservoir is *readable by ordinary chemistry*: the time-resolved yield of one product pool gives an out-of-sample information-processing capacity of 2.0, and the nuclear polarisation the product carries away raises it to 4.7, with no spectroscopy whatever. Second, there is no quantum advantage: a classical network given the same number of readout features matches the five-spin reservoir. Third, and decisively, the memory horizon is set *not* by the microsecond radical-pair lifetime but by the turnover interval, because the nuclear register survives recombination in the diamagnetic product. A physiological 1–100 ms turnover changes the memory capacity by $< 0.2\%$ and places the horizon at 3–280 ms, inside the neural band. The binding constraint is therefore chemical, not temporal, and falsifiable: the *same* nuclear register must be reused from one turnover to the next, failing which the capacity collapses to a memoryless read-back. This replaces a timescale argument, which needs no quantum mechanics, with a criterion that does.

I. INTRODUCTION

The radical-pair mechanism (RPM) is the only experimentally established way in which a magnetic field as weak as the geomagnetic field ($\sim 50\ \mu$T) can change the outcome of a chemical reaction [1, 2, 16]. A spin-correlated pair born in a singlet electronic state undergoes coherent singlet–triplet (S–T) interconversion driven by hyperfine fields; because recombination is spin-selective, a field that re-tunes this interconversion alters the product yields. The mechanism underlies the leading model of avian magnetoreception, where the photoreceptor cryptochrome hosts a flavin–tryptophan radical pair [2–5, 12].

The success of the RPM has repeatedly motivated a bolder conjecture: that spin coherence in biomolecules might support not merely sensing but *information processing*—a "quantum brain" [6, 7], a proposal long contested on decoherence grounds [9] and part of the broader programme of quantum biology [10, 11]. A concrete, named instance is the *three-layer quantum-brain hypothesis* [19, 20], which posits a layered architecture—a long-lived nuclear-spin memory (Layer 1), a radical-pair read/write interface (Layer 2), and classical neuronal transduction (Layer 3)—and which, in its most recent form, already reframes the proposal from quantum *computation* to quantum *reservoir dynamics*. The present paper is the quantitative test of that architecture as a reservoir, and the bound it returns. A companion study from this program places the conjecture on quantitative footing from the sensing side. Solving the radical-pair master equation with electronic decoherence explicitly included, it finds that the active-site radical pairs of the principal neurotransmitter-metabolising flavoenzymes (monoamine oxidase A/B, D-amino-acid oxidase) are magnetically inert ($< 10^{-4}\%$ field effect): their large exchange coupling ($|J| \sim 0.75$ GHz) and picosecond lifetimes each independently forbid weak-field S–T mixing. Only a *spatially separated* radical pair, as in cryptochrome—with $J \approx 0$ and a microsecond lifetime—sustains the coherence required for weak-field sensitivity [23, 24]. This premise is a standard result of radical-pair compass theory [2, 23] and is logically independent of the companion analysis; we take the existence of a long-coherence separated radical pair as our self-contained starting point.

This sharp dichotomy invites the question we address here. The separated radical pair is singled out because it preserves long-lived electron–nuclear spin coherence. *Is that coherence good for anything besides sensing the geomagnetic field?* We argue the question is best posed not as gate-model quantum computation but as *reservoir* computing, and we answer it with the chemistry left in: driven, read and clocked as a turnover actually is. We state at the outset the load-bearing caveat: whether a *neuronal* cryptochrome forms a light- or reaction-driven separated radical pair at all is unestablished—the mammalian CRY1/CRY2 proteins are primarily circadian transcriptional regulators [15]—so every biological conclusion below is conditional on the existence of such a pair. The spin-dynamics results (capacity, classical baselines, the register criterion) do not depend on that condition and stand on their own.

Why a reservoir, not a gate. Gate-model quantum computation treats decoherence as the adversary and demands error correction; a warm, wet, 10^{10}-copy biomolecular ensemble cannot meet that bar. Reservoir computing (RC) inverts the relationship. An RC system feeds a time-dependent input into a fixed nonlinear dynamical "reservoir" and trains only a *linear* readout of its state [25, 26]. RC *requires* dissipation: the echo-state / fading-memory property—that the influence of past inputs decays—is supplied precisely by relaxation [27, 28]. The molecular-spin and NMR communities have demonstrated quantum RC explicitly in nuclear- and electron-spin ensembles [27–29], part of a broad physical-reservoir-computing effort [13]. In this system the fading memory is supplied not by electronic dephasing—removing it entirely leaves the capacity and the memory kernel essentially unchanged—but by the turnover itself: each cycle discards the electron pair and keeps only the nuclei, which is a dissipative, non-unitary map. It is the openness of the chemical cycle, not the decoherence of the electrons, that makes the system a reservoir.

Our contribution is to answer that question with the chemistry left in, and the answer overturns three natural expectations. (i) We drive the reservoir as a real turnover does—nuclei retained, a new spin-correlated pair created each cycle—rather than by resetting a single electron, and read it through the products a downstream reaction could actually use. (ii) We compare against classical reservoirs matched by readout dimension, and find no quantum advantage, which we report rather than argue around. (iii) We show that the timescale objection usually raised against this system rests on identifying the reservoir's clock with the radical-pair lifetime, and replace it with a chemical criterion—reuse of the nuclear register—that is sharper, falsifiable, and, unlike a comparison of two timescales, actually requires the quantum treatment.

The physical correspondence we adopt, adapting the spin-RC framework of Fujii and Nakajima [27], maps the reservoir-computing elements onto the radical pair as follows: the long-T_2 memory nodes are the nuclear spins (^{31}P, ^{1}H); the nonlinear read/write head is the radical-pair electron system; the read/write channel is the hyperfine coupling; and the fading memory (echo-state property) is supplied by the electronic T_2^e together with inter-cycle nuclear dephasing. These are exactly the three layers of the hypothesis above: Layer 1 (nuclear-spin memory), Layer 2 (the radical-pair interface), and Layer 3 (the classical readout/transduction that trains the linear map).

The paper is organized as follows. Section II establishes the reservoir's capacity and its convergence, localises that capacity in spin coherence, compares it against matched classical baselines, and then asks what a chemical readout recovers; it closes with the reservoir's clock and the nuclear-register criterion. Section III places the result among quantum-brain proposals and states the limitations. The model, protocol, and capacity definitions are collected in Appendix A.

II. RESULTS

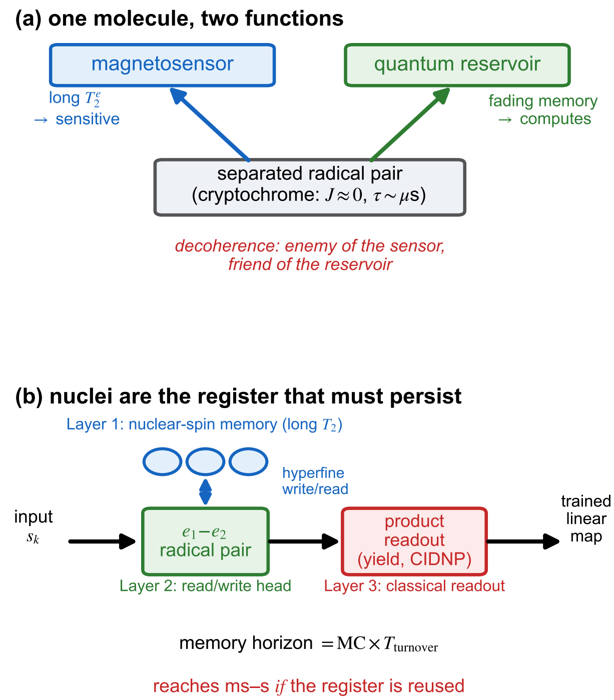

FIG. 1. **The reservoir and the quantity that decides its reach.** (a) A cryptochrome-type separated radical pair ($J \approx 0$, lifetime $\tau \sim \mu$s) is the same molecule in two roles: long electronic coherence makes it a weak-field magnetosensor, while the dissipation of an open cycle supplies the fading memory a reservoir needs. (b) The nuclei are the register: they are not consumed by recombination and therefore persist from one turnover to the next, whereas the electron pair is created afresh each cycle. The memory horizon is consequently $\mathrm{MC} \times T_{\mathrm{turnover}}$, not $\mathrm{MC} \times \tau$—and it reaches the neural band only if that register is genuinely reused.

We drive the reservoir the way a real turnover does. Each cycle the nuclei are retained and a *new* spin-correlated electron pair is created whose singlet character encodes the input, $\rho_e(s) = s\,|S\rangle\langle S| + (1-s)P_T/3$; the pair then evolves under the Liouville-space master equation with electronic dephasing and, where the readout requires it, spin-selective recombination. This replaces the reset of a single electron used in earlier treatments, which is not a chemical process and which—for a separated pair—destroys the electron–electron correlation every cycle. All capacities are out-of-sample (trained on the first half of a run, scored on the held-out second half) and are means over 6–8 input realisations. Model, protocol and estimator are given in Appendix A.

A. Capacity of the reservoir

With a twelve-observable readout the five-spin reservoir attains

$$\mathrm{MC} = 3.9, \qquad \mathrm{IPC} = 5.6 \pm 0.2,$$

of which 1.7 units are nonlinear. Unlike the single-electron-reset protocol, the estimate is converged in the length of the driving sequence: repeating at $L = 600, 1200, 2400, 4800$ gives $\mathrm{IPC} = 5.28, 5.60, 5.87, 5.84$, i.e. flat to within the seed-to-seed scatter beyond $L \approx 2400$. The memory kernel decays over ~ 4 cycles, the signature of the fading-memory property.

B. The capacity is largely coherent

To ask how much of this is quantum we add an all-spin pure-dephasing rate γ in the S_z product basis and take the incoherent limit at *matched hopping rate*: since the golden-rule transfer rate scales as $|H_{ij}|^2/\gamma$, holding rate $\times\, \tau$ fixed requires $\tau \propto \gamma$. Doing so, the capacity saturates at $\mathrm{IPC} = 1.00$ for $\gamma = 10^3, 5\times10^3, 2\times10^4$ MHz — exactly the memoryless floor, i.e. instantaneous read-back of the input with no memory at all. Against the coherent value 5.42 this is a coherence fraction

$$\frac{\mathrm{IPC}_{\gamma=0} - \mathrm{IPC}_{\mathrm{incoh}}}{\mathrm{IPC}_{\gamma=0}} = 0.82. \qquad (1)$$

Taking the limit at fixed τ instead, as is sometimes done, measures Zeno freezing rather than the classical limit and inflates this number; we avoid that.

C. There is no quantum advantage

A capacity is only meaningful against a classical reference, and the comparison must be symmetric in what is read out. Driving tanh echo-state networks (ESNs) with the same input and scoring them with the *same* estimator:

reservoir (readout features)	IPC
quantum, 5 spins (12 observables)	5.59 ± 0.19
ESN, 5 nodes (5 linear features)	4.55 ± 0.32
ESN, 5 nodes (12 features: 5 linear +7 quadratic)	9.41 ± 0.86
ESN, 12 nodes (12 linear features)	10.37 ± 0.59

Matched by *physical unit* the quantum reservoir is modestly ahead (5.59 vs 4.55), but matched by *readout dimension*—the resource that actually sets the Dambre ceiling—a classical system of the same size is well ahead (9.41 vs 5.59). We therefore make no claim of quantum advantage. Which accounting is the fair one depends on a cost model for non-commuting measurements versus multiplications that we do not have; we report the numbers and leave the claim unmade.

D. Biology can read the reservoir

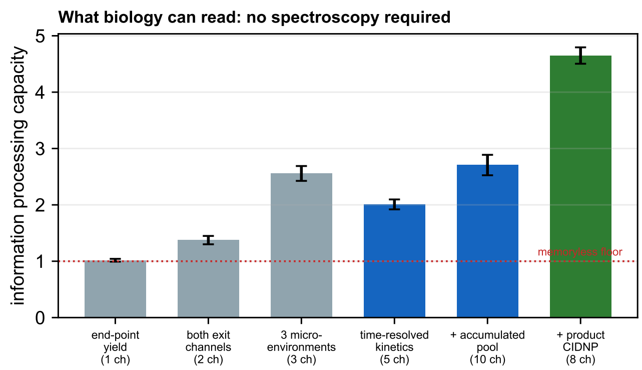

FIG. 2. **Readout is not the bottleneck.** Capacity recovered by chemically accessible observables at the cryptochrome operating point ($B = 50\ \mu$T, $J = 0$, $\tau = 1\ \mu$s), with spin-selective recombination switched on so that the observable is the product actually formed. The dotted line is the memoryless floor ($\mathrm{IPC} = 1$): a readout at that level reports the current input and nothing else. Ordinary reaction kinetics clear it; adding the nuclear polarisation carried away by the product (CIDNP) more than doubles the capacity again. No spectroscopy is involved.

Biology has no EPR spectrometer; it has products. We therefore restrict the readout to observables a downstream reaction could plausibly use, with recombination switched on so that the quantity read is the yield actually produced, $Y_X = k_X \int_0^\tau \mathrm{Tr}[P_X \rho(t)]\, dt$ (Fig. 2):

readout route	channels	IPC
end-point singlet yield	1	1.02
both exit channels, end point	2	1.37
three micro-environments (different τ)	3	2.56
time-resolved kinetics of one product pool	5	2.01
+ accumulated downstream pool	10	2.71
+ CIDNP polarisation on the product	8	**4.65**

Two things follow. First, the readout is not the limiting resource: time-resolved product kinetics alone clear the memoryless floor by a factor of two, and the nuclear polarisation the product carries away raises the capacity to 4.65 without any spectroscopic access whatsoever. Second, a single scalar end-point yield gives $\mathrm{IPC} = 1.02$, which saturates its own one-observable bound but consists almost entirely of the zero-delay term: it reports the present input and remembers nothing. Claims about reservoir behaviour should not be based on it.

The mechanism behind the CIDNP column is worth stating, because it also explains a negative result. The reduced state of *either* electron of a newly born pair is maximally mixed for every input, since both $|S\rangle\langle S|$ and $P_T/3$ have zero one-electron polarisation; the input lives entirely in the two-electron correlation. With $J = 0$ the two radicals do not interact, so nothing can transfer that correlation to a nucleus—and indeed, with recombination switched off, the cryptochrome-point capacity is 1.01, the memoryless floor. It is *spin-selective recombination* that converts the singlet–triplet correlation into a population difference and so writes the input into the nuclear register. In a separated radical pair, recombination is not merely how the reservoir is read; it is how it is written.

E. The clock is the turnover, not the pair lifetime

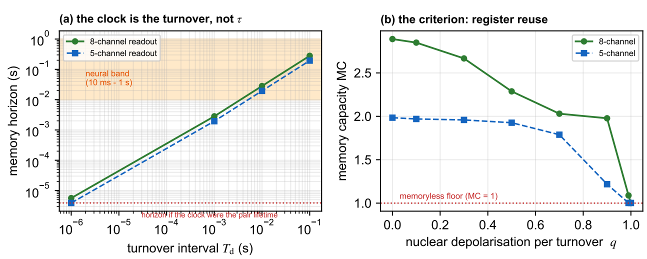

FIG. 3. **(a)** The memory horizon $\mathrm{MC} \times T_{\mathrm{turnover}}$ against the turnover interval. Because the nuclear register survives recombination, inserting a physiological pause leaves the memory capacity essentially unchanged and moves the horizon into the neural band (shaded). The dotted line is where the horizon would sit if the clock were the radical-pair lifetime. **(b)** The criterion. As the nuclear register is depolarised between turnovers with probability q, the capacity falls to the memoryless floor: at $q = 1$—a fresh molecule each cycle—MC = 1.00 and there is no memory.

The horizon over which a reservoir remembers is MC multiplied by its cycle time. It is natural, and it is what the submitted version of this work did, to identify that cycle time with the radical-pair lifetime $\tau \sim 1\ \mu$s and conclude that a 3 μs horizon puts the system 10^3–10^5 below neural timescales. That identification is wrong, and its own model says so. The carrier of the memory is the nuclear register, and nuclei are not consumed by recombination: they leave in the diamagnetic product, where nuclear relaxation times are of order 0.1–1 s. The cycle time of the reservoir is therefore the interval between turnovers.

Inserting a pause T_d between turnovers and relaxing the register with probability $q = 1 - \exp(-T_\mathrm{d}/T_{1n})$ (with $T_{1n} = 1$ s) gives Fig. 3(a):

T_d	q	MC (8 ch)	horizon
1 μs	10^{-6}	2.818	5.6 μs
1 ms	10^{-3}	2.818	2.8 ms
10 ms	10^{-2}	2.817	28 ms
100 ms	9.5×10^{-2}	2.800	280 ms

The memory capacity changes by 0.2% across five orders of magnitude in the turnover interval, so the horizon simply tracks it, and a perfectly ordinary enzymatic or photocycle turnover of 1–100 ms places it squarely in the neural band. The timescale argument does not survive; the same spin physics runs on a slow clock.

F. The criterion: reuse of the nuclear register

What replaces it is a chemical requirement, and a sharper one. Everything above assumes that the register the reservoir writes into is the *same* register on the next turnover. Relaxing that assumption by depolarising the nuclei with probability q between cycles gives Fig. 3(b): memory is essentially intact for $q \lesssim 0.5$ (that is, while the turnover interval is shorter than the nuclear relaxation time), degrades through $q \approx 0.7$–0.9, and at $q = 1$—a fresh molecule every cycle—collapses to MC = 1.00, the memoryless floor, for every readout we tested.

This is the constraint that actually decides whether a radical pair can serve as a biological reservoir:

> *The same nuclear register must be re-used across successive turnovers, and successive turnovers must be driven by distinct input samples. Given that, the memory horizon is* $\mathrm{MC} \times T_{\mathrm{turnover}}$*, bounded above by* $\mathrm{MC} \times T_{1n} \sim 1$ *s; without it, the system retains nothing.*

Unlike a comparison of two timescales—which requires no quantum mechanics at all—this criterion is a statement about a spin register and can only be evaluated by the kind of analysis performed here. It is also directly falsifiable by chemistry: it fails if the radical pair is regenerated on a fresh molecule each turnover, or if the relevant nuclear relaxation in the diamagnetic intermediate is faster than the turnover rate.

III. DISCUSSION

What changed, and why it matters. Three claims that are natural to make about this system—that a scalar chemical readout is too impoverished to be useful, that a spin reservoir of a few spins beats a classical system of the same size, and that microsecond radical-pair lifetimes put biological computation out of reach—all fail when the model is driven as a real turnover and compared symmetrically. The readout is not the bottleneck; the quantum advantage is not there; and the timescale gap is an artifact of identifying the reservoir clock with the pair lifetime rather than the turnover interval. What remains is a single, sharp, chemical criterion.

Relation to prior work. Quantum reservoir computing in spin ensembles is established, including the use of dissipation as a resource and the exponential-state-space argument.[13, 27–30] We do not claim these as new, and—having found no advantage under a symmetric comparison—we do not add to them. Our contribution is to take a chemically realised radical pair, drive and read it as chemistry actually would, and convert a long-standing qualitative conjecture[6, 7, 9] into a criterion that experiment can test. The result also sharpens why CIDNP matters here: in a separated pair it is not an optional extra channel but the mechanism by which information reaches the nuclei at all.

Limitations. The model is a five-spin reduction with isotropic hyperfine couplings; the full anisotropic tensor set of the cryptochrome pair[21, 22] is not treated, and ^{14}N is modelled as spin-1/2. The capacity estimator uses degree-2 targets, so higher-order nonlinear capacity is not counted. Whether a *neuronal* cryptochrome forms a separated radical pair at all is unestablished—CRY1/CRY2 are primarily circadian transcriptional regulators[15]—so the biological reading of our criterion is conditional. The spin-dynamics results do not depend on that condition.

What would settle it. The criterion is addressable in vitro before it is addressable in a neuron. Covalently linked donor–bridge–acceptor radical pairs have synthetically tunable exchange coupling and lifetime and are read by transient EPR and optical/CIDNP spectroscopy.[2, 14, 34] Repeatedly cycling such a system and asking whether the product yield of turnover k depends on the input at turnover $k-1$, $k-2$, ... measures the memory horizon directly, and varying the interval between cycles tests the register-reuse criterion where it is sharpest.

IV. CONCLUSION

Driven and read as a chemical turnover rather than as an abstract quantum channel, a cryptochrome-type radical pair is a reservoir whose output ordinary chemistry can recover: time-resolved product kinetics give IPC = 2.0 and the nuclear polarisation carried off by the product raises this to 4.7, with no spectroscopic access required. Eighty-two per cent of the capacity is coherent in origin, and yet there is no quantum advantage—a classical network with the same number of readout features matches or exceeds it. The obstacle usually invoked, a microsecond memory horizon, does not survive scrutiny: the register is nuclear and survives recombination, so the clock is the turnover interval and a physiological 1–100 ms turnover places the horizon at 3–280 ms. What decides the question instead is whether the same nuclear register is reused from one turnover to the next; if it is not, the capacity falls to a memoryless MC = 1. We offer that criterion, rather than a timescale comparison, as the thing to test.

ACKNOWLEDGMENTS

We thank the reviewers of the companion sensing study, whose insistence on decoherence-explicit treatment shaped the methodology used here. All numerical results are reproducible from the open-source `qbscreen` package.

AUTHOR CONTRIBUTIONS

H.W. conceived the study, developed the master-equation and reservoir code, ran the simulations, and wrote the manuscript. T.T. contributed to the spin-dynamics formulation and the analysis of the nuclear-register criterion. Both authors reviewed the results and approved the manuscript.

Appendix A: Model and methods

1. Reservoir Hamiltonian

We model the reservoir as $N = 5$ spins—two radical-pair electrons (e_1, e_2) and three nuclei (^{31}P and two ^{1}H)—with Hamiltonian (in MHz)

$$H = \omega_e \left(S_z^{e_1} + S_z^{e_2}\right) + \sum_{\mu=x,y,z} \Big[A_P\, S_\mu^{e_1} I_\mu^{P} + A_{H_1} S_\mu^{e_1} I_\mu^{H_1} + A_{e_2} S_\mu^{e_2} I_\mu^{H_2} + J\, S_\mu^{e_1} S_\mu^{e_2}\Big], \tag{A1}$$

where $\omega_e = g_e \mu_B B/h$ is the electron Zeeman frequency. Each radical carries its own nuclei—they are not shared, which is what a spatially separated pair means—and the exchange J links the two electrons. At the engineered operating point we use $A_{e_1a} = 80$, $A_{e_1b} = 40$, $A_{e_2a} = 15$ MHz, $J = 2$ MHz and $B = 1$ mT: representative molecular-spin-RC values chosen to give effective hyperfine writing, not a fit to a specific protein. For the cryptochrome point we set $J = 0$, $B = 50$ μT and $\tau = 1$ μs, with hyperfine magnitudes representative of the dominant FAD$^{\bullet-}$ and TrpH$^{\bullet+}$ couplings [21, 22]. The conclusions are not fine-tuned to these numbers: the capacity varies by less than $\sim$ 20% over an order-of-magnitude scan of T_2^e, and the register criterion of Sec. II is independent of the hyperfine assignment.

2. Master equation and propagator

Each reservoir update is the completely positive, trace-preserving map $\rho \mapsto e^{\mathcal{L}\tau}\rho$ generated by the Haberkorn master equation [31] (whose spin-selective recombination operator we adopt in its standard form [17]) with electronic pure dephasing,

$$\frac{d\rho}{dt} = -i[H,\rho] - \tfrac{1}{2}\{k_S P_S + k_T P_T, \rho\} + \sum_i \frac{1}{T_2^e}\left(S_z^i \rho S_z^i - \tfrac{1}{4}\rho\right), \tag{A2}$$

solved exactly in Liouville space. The third term—the electronic decoherence that the companion sensing analysis showed to be decisive—is here the source of fading memory. We verified that, in the limit $1/T_2^e \to 0$ and $k_S = k_T$, the Liouville-space solver reproduces the closed-form coherent Green's-function singlet yield to machine precision (Supplemental Material). Within an update step no recombination acts ($k_S = k_T = 0$); the readout step uses the spin-selective term.

3. Reservoir protocol

We use the standard temporal-QRC protocol [27]. At step k the input electron is reset to the input-encoded pure state $|\psi(s_k)\rangle = \sqrt{1-s_k}\,|0\rangle + \sqrt{s_k}\,|1\rangle$ while the reservoir (the remaining electron plus nuclei) is retained; the full state evolves for a fixed lifetime τ under $e^{\mathcal{L}\tau}$ ($\tau = 50$ ns, $T_2^e = 50$ ns at the operating point); a set of observables is read out, and a linear (ridge) readout is trained on them. Inputs s_k are i.i.d. uniform on $[0,1]$; the first ~ 100 steps are discarded as washout.

This protocol has a natural chemical reading that addresses its physical realizability. The "reset" of the input electron is not an applied operation but the *birth of a new radical pair*: each catalytic or photochemical cycle creates a fresh spin-correlated electron pair whose initial state encodes the instantaneous input (e.g. a substrate-flux- or light-dependent singlet/triplet bias), while the surrounding nuclei—which are not consumed by recombination—carry spin polarisation across cycles. The reservoir is thus a *persistent nuclear bath repeatedly interrogated by transient electron pairs*, which is precisely the radical-pair situation, not an artificial gate sequence. The washout discards initial transients, so the reported capacities are independent of the initial spin state—a direct consequence of the fading-memory (echo-state) property and the reason no thermal-population fine-tuning is needed.

4. Capacity measures

We quantify the reservoir with two information-theoretic benchmarks. The linear *memory capacity* [32] $\mathrm{MC} = \sum_{d\geq 0} r^2(s_{k-d})$ sums the squared multiple correlation for reconstructing delayed inputs. The *information processing capacity* [33] adds nonlinear functions of the input history, $\mathrm{IPC} = \mathrm{MC} + \sum_{\mathrm{nonlinear}} C$, evaluated here through degree-2 Legendre targets (diagonal $P_2(s_{k-d})$ and cross $s_{k-d_1}s_{k-d_2}$ terms). A fundamental theorem [33] bounds the total capacity by the number of linearly independent readout observables, $\mathrm{IPC} \leq n_{\mathrm{obs}}$; we use this bound throughout as a correctness anchor and as the key to interpreting the ensemble results.

All results are reproducible from the open `qbscreen` package (`reservoir.py`, `ensemble.py`, `quantum_vs_classical.py`; 28/28 tests pass).

Appendix B: Data and code availability

All analyses are implemented in the open `qbscreen` package, `https://github.com/deeptell-inc/new_Q_brain`. The corrected correlated-pair-birth injection is in `qbscreen/corrected_injection.py`; the chemically accessible readout routes, the turnover-clock scan and the register-reuse criterion are in `qbscreen/readout_routes.py`; the protocol-consistent capacity, coherence fraction and classical baselines are in `qbscreen/final_numbers.py`; the master-equation solver is in `qbscreen/master_equation.py`. Numerical outputs are in `simulation_results/`. The capacity bound and the agreement of the solver with the closed-form coherent limit are checked in `qbscreen/tests/` (28 tests, all passing).

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